# OCC: SOFTWARE ENGINEER YEAR 2

## SAQA ID

## NQF 6

## Software design and testing C#

## SDT621

## PRACTICAL MODULE -PM-04

## 2026

CTU training solutions | Private Tertiary Education

# Table of Contents

# DECLARATION OF AUTHENTICITY

I BELINDA MOKOENA (FULL NAME)

hereby declare that the contents of this assessment PM04

is entirely my own work, completed by me without any paraphrasing/copying, or presented as my own work accessed from any AI Apps, example ChatGPT, Co-pilot, Perplexity, or any other App.

I have read and understood the Examination Rules and Regulations.

Signature: B.Mokoena

Date: 24/05/26

# QUALIFICATION AND ASSESSMENT DETAILS

| | |
|---|---|
| **Programme Title** | OCC: SOFTWARE ENGINEER |
| **Module Title** | Software design and testing C# |
| **Module Code** | SDT621 |
| **NQF Level** | 6 |
| **Assessment Type** | PRACTICAL MODULE (PM-O4) |
| **Hand Out Date** | 04 / 0 5 / 2026 |
| **Hand in Date** | 15 / 05 / 2026 |
| **Total Marks** | 100 |
| **Project Lead Developer** | Lusukama Selemani |
| **Internal Moderator** | Faith Muwishi |
| **Internal Moderator** | Michelle Du Toit |
| **External Moderator** | |

| | |
|---|---|
| **Student Name and Surname** | |
| **Student Number** | |

## Instructions:

Please read the following instructions carefully before attempting this Practical Module (PM-04):

General Instructions

- Read each question carefully and take note of the mark allocation before answering.
- Ensure that all questions are answered.
- Provide clear, well-structured C# code and explanations where required.
- Apply your understanding of software testing and debugging concepts (PM-04) when solving the tasks.

## Submission Requirements

For your final submission, you must include the following:

- Declaration of Authenticity (signed)

  (If the declaration is already included in this document, you may complete and sign it within this document.)
- Completed Answer Sheet (PDF format)

  (Preferably use this current document as your answer sheet.)
- Visual Studio Project Files

  (Submit your complete project compressed as a ZIP file.)
- GitHub Repository Link

  (Ensure your repository contains the full and correct solution.)

## Evidence Collection Requirements

To support your submission, you must:

- Capture screenshots of the application output and testing results
- Ensure each screenshot displays the full screen, including system date and time
- Include evidence of:
  - Application running (GUI output)
  - Test results (MSTest / Test Explorer)
- Do NOT submit screenshots of source code
- Submit your complete project folder as a ZIP file

## GitHub Submission Requirement

You are required to:

- Push your project code to GitHub
- Ensure the repository contains the correct and complete solution
- Share the GitHub repository link in your submission for review and verification
- Example: https://github.com/your-username/project-name

## Testing Requirements (PM-04 Focus):

You are required to demonstrate:

- Unit Testing (MSTest for payroll calculations)
- Integration Testing (input → calculation → output)
- System Testing (complete application workflow)
- Error handling and validation testing
- Debugging and correction of defects

## Test Report Requirement

You must produce a **Test Report** that includes:

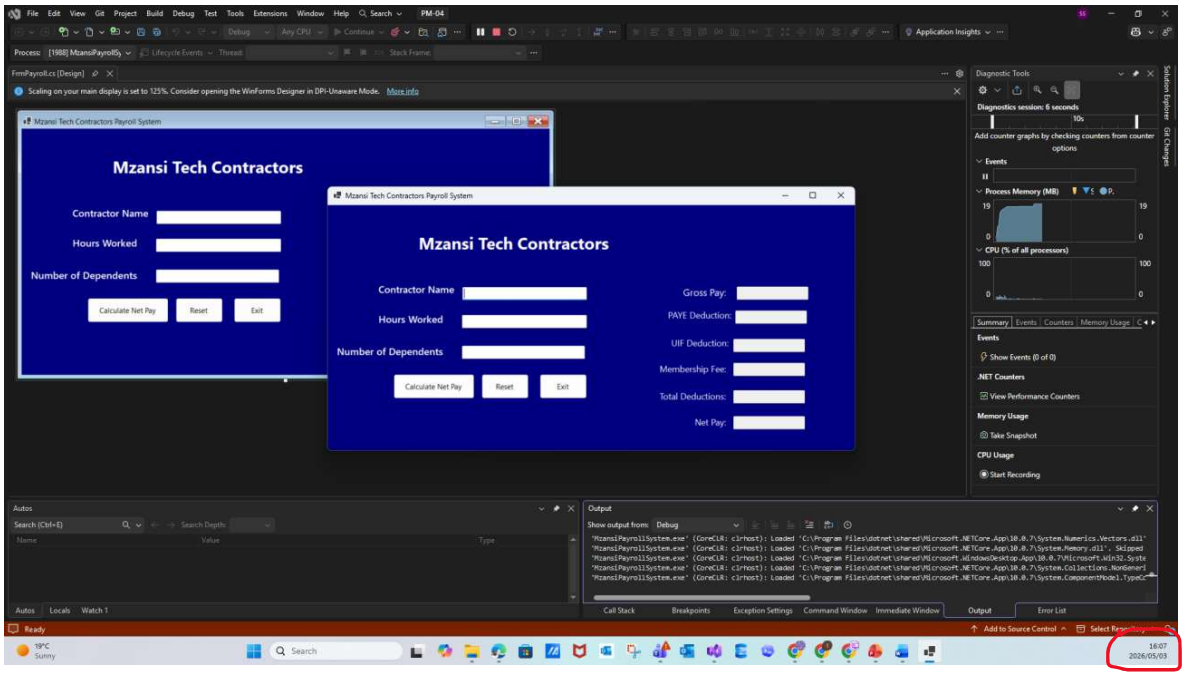
- Test cases
- Expected results
- Actual results
- Pass/Fail status
- Identified defects
- Corrections made
- Retesting results

# Assessment Outcomes:

- IAC0102 – The correct tests are applied
- IAC0103 – A test report is produced
- PM-04 – Testing and debugging software systems

## Example of Collecting Evidence for your submission pdf:

Mzansi Tech Contractors Payroll System Design :




Ensure each screenshot shows the **full screen**, including the **system date and time**

# Payroll Testing and Debugging Project

## Scenario: Mzansi Tech Contractors Payroll System

Mzansi Tech Contractors is a South African company that places temporary IT contractors, including systems analysts and software developers, with client companies. Contractors are paid in South African Rand (R) based on the number of hours worked.

The company uses a payroll application to calculate each contractor's gross pay, statutory deductions, membership deductions, and net pay. However, management has reported that the current system sometimes produces incorrect calculations and does not properly validate user input.

### Business Rules

The application must accept:

- Contractor name
- Number of hours worked
- Number of dependents

**The system must calculate:**

**Hourly rate**: R950.00 per hour

**Gross Pay** = Hours Worked × Hourly Rate

**Deductions:**

**UIF Deduction**: 1% of gross pay, based on South African UIF rules where the employee contributes 1% and the employer contributes 1%.

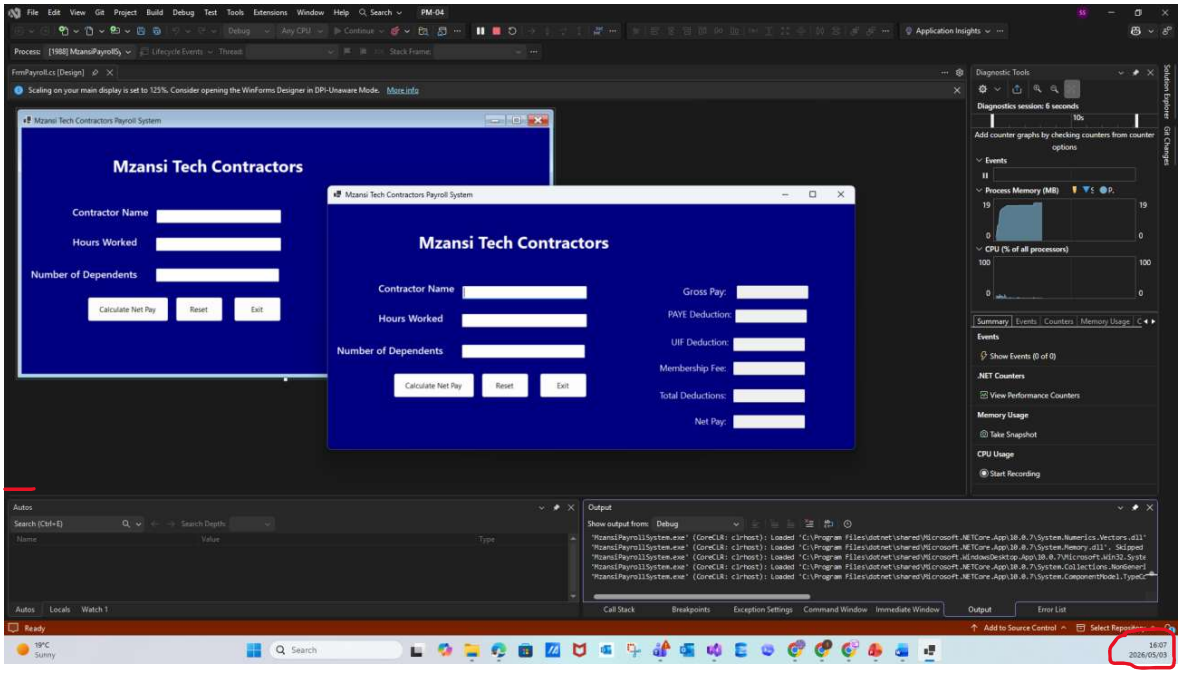
**PAYE Deduction**: Use a simplified SARS-based payroll rule for the assessment. SARS provides official employee tax deduction tables for PAYE calculations.

For this practical assessment, use:

- **PAYE** = (Gross Pay - (Gross Pay × 0.0575 × Number of Dependents)) × 25%
- **Membership Fee**: 13% of gross pay
- **Net Pay** = Gross Pay – UIF – PAYE – Membership Fee

# PM-04 Practical Requirement

Design a Windows Forms GUI Application that allows the user to capture contractor payroll details and display:




- Gross pay
- UIF deduction
- PAYE deduction
- Membership fee
- Total deductions
- Net pay

# Required Testing Levels for PM-04

Learners must perform the following testing levels:

## 1. Unit Testing

Test individual calculations separately.

| Test Case | Input | Expected Output | Result |
|---|---|---|---|
| Gross Pay | | | |
| UIF | | | |
| Membership | | | |
| PAYE | | | |
| Net Pay | | | |

## 2. Integration Testing

Test whether the input, calculation, and output sections work together correctly.

Examples:

Hours entered by the user are correctly used in the gross pay calculation.

Gross pay is correctly passed into UIF, PAYE, and membership calculations.

All deductions are correctly combined to calculate net pay.

### 3. System Testing

Test the full payroll application as one complete system.

| Step | Action | Expected Result |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |

### 4. Retesting

After fixing identified defects, learners must run the same failed tests again to confirm that the system now works correctly.

### 5. Regression Testing

Learners must confirm that fixing one defect did not break another part of the payroll application.

Produce a short test report showing expected results, actual results, defects found, and corrections made

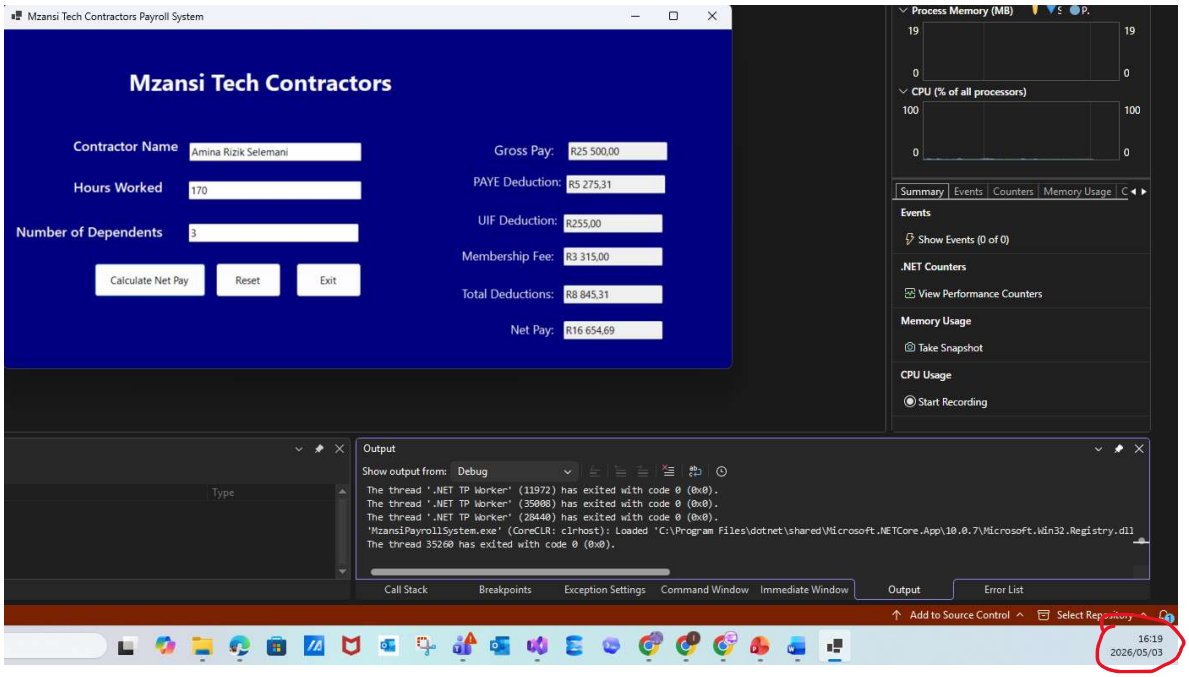
## Validation Rules

The system must not allow:

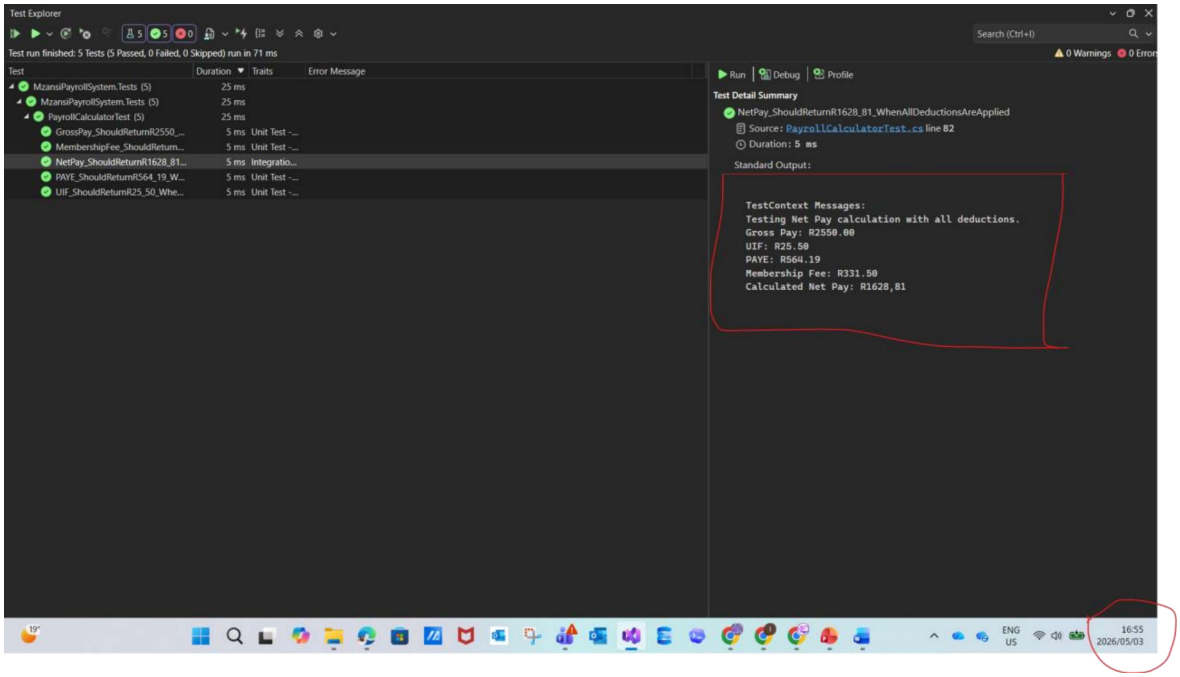
- Empty contractor name
- Negative hours worked
- Negative dependents
- Non-numeric input for hours or dependents
- Dependents greater than a reasonable limit, for example 10

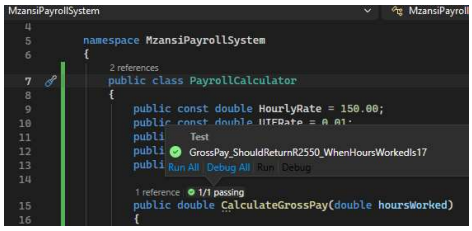
**END OF PAPER**

## Output Example




## After test passed:



# PM-04 Marking Rubric

| Section | Criteria | Marks |
|---|---|---|
| **User Interface Design (10)** | Clear and professional layout | 2 |
| | Correct labels and input fields | 2 |
| | Proper alignment and spacing | 2 |
| | Appropriate use of controls | 2 |
| | Output section clearly structured | 2 |
| **Input Validation (10)** | Validates empty fields | 2 |
| | Validates numeric input | 2 |
| | Prevents negative values | 2 |
| | Validates dependents range (0–10) | 2 |
| | Displays user-friendly error messages | 2 |
| **Payroll Calculations (20)** | Correct Gross Pay calculation | 4 |
| | Correct UIF calculation (1%) | 4 |
| | Correct Membership Fee (13%) | 4 |
| | Correct PAYE calculation | 4 |
| | Correct Net Pay calculation | 4 |
| **Application Functionality (10)** | Calculate button works | 3 |
| | Reset button clears fields | 3 |
| | Exit button works | 2 |
| | System runs without errors | 2 |
| **Unit Testing (20)** | Test project created correctly | 3 |
| | PayrollCalculator class used | 3 |
| | Gross Pay test | 3 |
| | UIF test | 3 |
| | Membership Fee test | 3 |
| | PAYE test | 3 |
| | Net Pay test | 2 |
| **Test Quality & Documentation (15)** | Meaningful test names | 3 |
| | Use of TestCategory | 3 |
| | Use of TestContext logging | 3 |
| | Expected vs actual validation | 3 |

| | | |
|---|---|---|
| | Professional structure | 3 |
| **Integration & System Testing (10)** | Integration testing explained | 3 |
| | System testing scenario | 3 |
| | End-to-end workflow tested | 2 |
| | Evidence provided (screenshots/logs) | 2 |
| **Code Quality (5)** | Clean, readable code | 2 |
| | Proper naming conventions | 2 |
| | Separation of logic (Calculator class) | 1 |
| **TOTAL** | | **100** |

# PM-04 Mapping: Mzansi Tech Contractors Payroll System

| PM-04 Requirement / Outcome | Project Activity | Evidence Required | Marks |
|---|---|---|---|
| Apply correct tests | Unit testing using MSTest for gross pay, UIF, PAYE, membership fee, and net pay | Test Explorer screenshot showing passed tests | 15 |
| Test system calculations | Verify all payroll formulas produce correct results | Test cases with expected vs actual results | 15 |
| Test integrated components | Confirm inputs, calculation class, and output textboxes work together | Screenshot of completed payroll form | 10 |
| Perform system testing | Test the full application from start to finish | System test table and screenshots | 10 |
| Identify defects | Record any errors found during testing, such as wrong formula or failed reset button | Defect log / bug report | 10 |
| Debug source code | Correct identified errors in the application | Before-and-after explanation or screenshots | 10 |
| Retest fixed defects | Run the same test again after fixing the issue | Retest evidence showing pass result | 10 |
| Produce test report | Document test cases, expected results, actual results, and pass/fail status | Completed test report | 10 |
| Follow coding standards | Use clear names, readable code, and separated PayrollCalculator class | Source code screenshots / GitHub link | 5 |
| Provide complete project evidence | Submit application, tests, screenshots, and report | Final zipped project / GitHub repository | 5 |
| **Total** | | | **100** |